ACM/IDS 104 APPLIED LINEAR ALGEBRA
PROBLEM SET 5: SOLUTION

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Problem 1.(10 points) Application of Projections to Approximation
Complete Problem 1 in `PS5.mlx`.

Solution: See `PS5solutions.mlx` (uploaded to the Piazza page). The interpolating and approximating polynomials are shown in Figure 1.

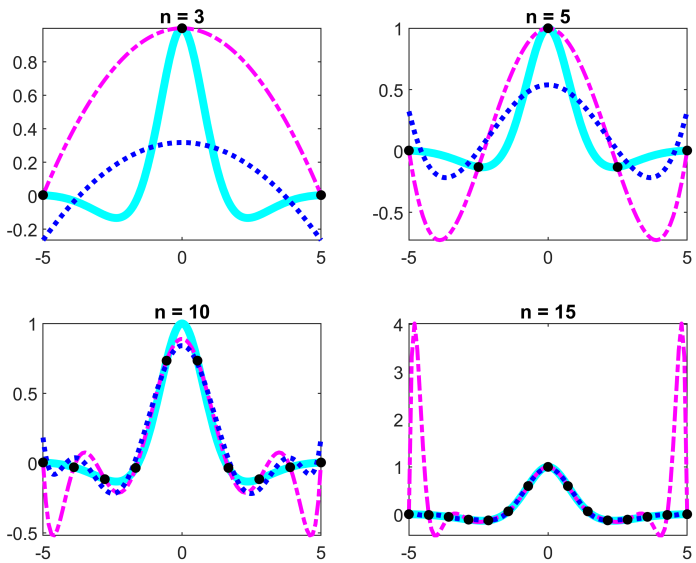

Figure 1. Approximation (dark blue) vs Interpolation (pink or purple?).

Unlike interpolating polynomials, approximating polynomials do not exhibit Runge's phenomenon (large oscillations between data points).

Problem 2. (10 points) Hermite Polynomials
Consider the following inner product, sometimes called the Hermite inner product[1]:

$$\langle f, g\rangle = \int_{-\infty}^{\infty} f(x)g(x)e^{-\frac{x^2}{2}}\,dx. \tag{1}$$

This inner product appears in probability theory, functional analysis, numerical PDEs, and in physics[2].

Let $f(x) = e^x$, and suppose that we want to approximate this function by a polynomial $p(x) \in \mathcal{P}^{(n)}(\mathbb{R})$ of degree at most n on the real line. Notice that we cannot do this with respect to the

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[1]Technically, this inner product is defined on the Hilbert space $L^2(\mathbb{R}, \mu)$, where the measure $d\mu(x) = e^{-\frac{x^2}{2}}\,dx$, which consists of all (measurable) functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_{-\infty}^{\infty} f(x)^2 e^{-\frac{x^2}{2}}\,dx < \infty$. But for the purpose of this problem, this technicality is not needed. We consider only those functions f and g for which the integral in (1) exists.

[2]In physics, the weight e^{-x^2} is typically used instead of $e^{-\frac{x^2}{2}}$.

standard L^2 inner product (convince yourself that this is the case)[3]. We can do this, however, with respect to the Hermite inner product. In what follows, you can use Wolfram Alpha (or other tools) to compute the arising indefinite integrals.

(a) (3 points) Starting from the standard basis $1, x, x^2$ of the vector space $\mathcal{P}^{(2)}(\mathbb{R})$, use the Gram-Schmidt process to construct an orthogonal basis $h_0(x), h_1(x), h_2(x)$ of $\mathcal{P}^{(2)}(\mathbb{R})$ with respect to the Hermite inner product. The polynomials in this basis are called the *Hermite polynomials*[4].

(b) (3 points) Based on $h_0(x), h_1(x), h_2(x)$ construct an orthonormal basis $H_0(x), H_1(x), H_2(x)$ of $\mathcal{P}^{(2)}(\mathbb{R})$.

(c) (4 points) Find the best approximation $p(x) \in \mathcal{P}^{(2)}(\mathbb{R})$ to $f(x) = e^x$ in the Hermite sense, that is, with respect to the Hermite norm, induced by the Hermite inner product.

Solution:

(a) Applying the Gram-Schmidt process to the standard basis, we have:

$$\begin{aligned}
h_0(x) &= 1, \\
h_1(x) &= x - \frac{\langle x, 1\rangle}{\|1\|^2}1 = x - \frac{\int_{-\infty}^{\infty} x e^{-\frac{x^2}{2}}\,dx}{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx} = x - \frac{0}{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx} = x, \\
h_2(x) &= x^2 - \frac{\langle x^2, 1\rangle}{\|1\|^2}1 - \frac{\langle x^2, x\rangle}{\|x\|^2}x = x^2 - \frac{\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}}\,dx}{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx} - \frac{\int_{-\infty}^{\infty} x^3 e^{-\frac{x^2}{2}}\,dx}{\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}}\,dx}x \\
&= x^2 - \frac{\sqrt{2\pi}}{\sqrt{2\pi}} - \frac{0}{\sqrt{2\pi}}x = x^2 - 1.
\end{aligned} \tag{2}$$

(b) The corresponding orthonormal basis is

$$\begin{aligned}
H_0(x) &= \frac{h_0(x)}{\|h_0\|} = \frac{1}{\sqrt{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}}\,dx}} = \frac{1}{(2\pi)^{\frac{1}{4}}}, \\
H_1(x) &= \frac{h_1(x)}{\|h_1\|} = \frac{x}{\sqrt{\int_{-\infty}^{\infty} x^2 e^{-\frac{x^2}{2}}\,dx}} = \frac{x}{(2\pi)^{\frac{1}{4}}}, \\
H_2(x) &= \frac{h_2(x)}{\|h_2\|} = \frac{x^2 - 1}{\sqrt{\int_{-\infty}^{\infty} (x^2-1)^2 e^{-\frac{x^2}{2}}\,dx}} = \frac{x^2 - 1}{(8\pi)^{\frac{1}{4}}}.
\end{aligned} \tag{3}$$

(c) The best approximation $p(x) \in \mathcal{P}^{(2)}(\mathbb{R})$ to $f(x) = e^x$ is the closest (with respect to the Hermite norm) in $\mathcal{P}^{(2)}(\mathbb{R})$ point to $f(x)$. In other words, it is the orthogonal projection of $f(x)$ onto $\mathcal{P}^{(2)}(\mathbb{R})$:

$$p(x) = \mathrm{pr}_{\mathcal{P}^{(2)}(\mathbb{R})} f(x) = \alpha_0 H_0(x) + \alpha_1 H_1(x) + \alpha_2 H_2(x), \tag{4}$$

where the coordinates are given by the inner products:

$$\begin{aligned}
\alpha_0 &= \langle f, H_0\rangle = \frac{1}{(2\pi)^{\frac{1}{4}}} \int_{-\infty}^{\infty} e^x e^{-\frac{x^2}{2}}\,dx = \frac{(2e\pi)^{\frac{1}{2}}}{(2\pi)^{\frac{1}{4}}} = e^{\frac{1}{2}}(2\pi)^{\frac{1}{4}}, \\
\alpha_1 &= \langle f, H_1\rangle = \frac{1}{(2\pi)^{\frac{1}{4}}} \int_{-\infty}^{\infty} x e^x e^{-\frac{x^2}{2}}\,dx = \frac{(2e\pi)^{\frac{1}{2}}}{(2\pi)^{\frac{1}{4}}} = e^{\frac{1}{2}}(2\pi)^{\frac{1}{4}}, \\
\alpha_2 &= \langle f, H_2\rangle = \frac{1}{(8\pi)^{\frac{1}{4}}} \int_{-\infty}^{\infty} (x^2-1) e^x e^{-\frac{x^2}{2}}\,dx = \frac{(2e\pi)^{\frac{1}{2}}}{(8\pi)^{\frac{1}{4}}} = e^{\frac{1}{2}}\left(\frac{\pi}{2}\right)^{\frac{1}{4}}.
\end{aligned} \tag{5}$$

[3]We can do this only on the finite interval $[-a, a]$, see Practice Problem 10B.

[4]Hermite polynomials were originally defined by Laplace and studied in detail by Chebyshov. This is yet another manifestation of the Arnold principle: if a notion bears a personal name, then this name is not the name of the discoverer, `https://dsweb.siam.org/The-Magazine/All-Issues/vi-arnold-on-teaching-mathematics`.

Finally, combining (4) and (5) together, we have:

$$\begin{aligned} p(x) &= e^{\frac{1}{2}}(2\pi)^{\frac{1}{4}}\frac{1}{(2\pi)^{\frac{1}{4}}} + e^{\frac{1}{2}}(2\pi)^{\frac{1}{4}}\frac{x}{(2\pi)^{\frac{1}{4}}} + e^{\frac{1}{2}}\left(\frac{\pi}{2}\right)^{\frac{1}{4}}\frac{x^2-1}{(8\pi)^{\frac{1}{4}}} \\ &= e^{\frac{1}{2}}\left(1 + x + \frac{1}{2}(x^2-1)\right) = \frac{\sqrt{e}}{2}(x^2+2x+1) = \frac{\sqrt{e}}{2}(x+1)^2. \end{aligned} \tag{6}$$

Problem 3. (10 POINTS) ORTHOGONAL COMPLEMENTS
In this problem, we will practice finding orthogonal complements in different inner product spaces.

(a) (4 points) Let $W \subset \mathbb{R}^3$ be a subspace spanned by $v_1 = (1,2,3)^T$ and $v_2 = (2,0,1)^T$. Find the orthogonal complement
  (a1) (2 points) $W_1^\perp$ to W with respect to the dot product $\langle x, y\rangle_1 = x_1y_1 + x_2y_2 + x_3y_3$ and
  (a2) (2 points) $W_2^\perp$ to W with respect to the weighted product $\langle x, y\rangle_2 = x_1y_1 + 2x_2y_2 + 3x_3y_3$.

(b) (3 points) Let $U = \text{span}(I_n) \subset \mathbb{M}_{n\times n}$ be a subspace spanned by the identity matrix. Find the orthogonal complement $U^\perp$ to U with respect to the Frobenius inner product (introduced in Practice Problem 4B).

(c) (3 points) Let $V = \text{span}\{x, x^3\} \subset \mathcal{P}^{(3)}[-1,1]$ be a subspace of the inner product space $\mathcal{P}^{(3)}[-1,1]$ of polynomials on $[-1,1]$ of degree at most 3 equipped with the L^2 inner product. Find the orthogonal complement $V^\perp$.

Solution:
(a1) A vector $x = (x_1, x_2, x_3)^T \in W_1^\perp$ if and only if

$$\begin{aligned} \langle v_1, x\rangle_1 &= x_1 + 2x_2 + 3x_3 = 0, \quad \text{and} \\ \langle v_2, x\rangle_1 &= 2x_1 + x_3 = 0. \end{aligned} \tag{7}$$

Solving this system yields $x_2 = \frac{5}{2}x_1$ and $x_3 = -2x_1$. We therefore have

$$W_1^\perp = \text{span}\begin{bmatrix} 1 \\ \frac{5}{2} \\ -2 \end{bmatrix} \tag{8}$$

(a2) Similarly, $x = (x_1, x_2, x_3)^T \in W_2^\perp$ if and only if

$$\begin{aligned} \langle v_1, x\rangle_2 &= x_1 + 4x_2 + 9x_3 = 0, \quad \text{and} \\ \langle v_2, x\rangle_2 &= 2x_1 + 3x_3 = 0. \end{aligned} \tag{9}$$

Solving this system yields $x_2 = \frac{5}{4}x_1$ and $x_3 = -\frac{2}{3}x_1$. And, thus,

$$W_2^\perp = \text{span}\begin{bmatrix} 1 \\ \frac{5}{4} \\ -\frac{2}{3} \end{bmatrix} \tag{10}$$

(b) Recall that the Frobenius inner product $\langle A, B\rangle_F = \text{tr}(A^TB)$. Therefore, matrix $A \in U^\perp = (\text{span}(I_n))^\perp$ if and only if

$$\langle A, \alpha I_n\rangle_F = \alpha\,\text{tr}(A^T) = \alpha\,\text{tr}(A) = 0 \qquad \text{for any } \alpha \in \mathbb{R}. \tag{11}$$

Therefore, $U^\perp$ is the subspace of matrices with zero trace[5].

(c) Our goal is to find all polynomials $p(x) = ax^3 + bx^2 + cx + d$ such that $\langle p(x), x\rangle = 0$ and $\langle p(x), x^3\rangle = 0$. This leads to the following conditions on the coefficients:

$$\begin{aligned} \langle p(x), x\rangle &= \int_{-1}^{1}(ax^4 + bx^3 + cx^2 + dx)dx = \frac{2a}{5} + \frac{2c}{3} = 0, \\ \langle p(x), x^3\rangle &= \int_{-1}^{1}(ax^6 + bx^5 + cx^4 + dx^3)dx = \frac{2a}{7} + \frac{2c}{5} = 0. \end{aligned} \tag{12}$$

[5]Recall that back in Problem 1(b) of PS2, we established that the set of matrices with zero trace is indeed a subspace.

This system has only one solution: $a = c = 0$. Therefore,

$$V^{\perp} = \{p(x) = bx^2 + d, b, d \in \mathbb{R}\}. \tag{13}$$

Problem 4. (10 POINTS) REAL AND COMPLEX EIGENBASES
Consider the following matrix:

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \tag{14}$$

(a) (2 points) Is this matrix complete?
(b) (2 points) Does A admit an eigenbasis of $\mathbb{R}^3$? If yes, find it.
(c) (3 points) Does A admit an eigenbasis of $\mathbb{C}^3$? If yes, find it.
(d) (3 points) Construct a basis of $\mathbb{R}^3$ consisting of real and imaginary parts of eigenvectors of A. Is it an eigenbasis?

Solution:
(a) The characteristic polynomial of A is

$$p_A(\lambda) = \det(A - \lambda I) = \det \begin{bmatrix} -\lambda & 0 & -1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{bmatrix} = (1-\lambda) \det \begin{bmatrix} -\lambda & -1 \\ 1 & -\lambda \end{bmatrix} = (1-\lambda)(\lambda^2 + 1). \tag{15}$$

It has three roots (eigenvalues of A):

$$\lambda_1 = 1, \ \lambda_2 = i, \ \lambda_3 = \overline{\lambda_2} = -i. \tag{16}$$

Since all eigenvalues are distinct, all of them are automatically complete, since their algebraic and geometric multiplicities are both equal to one. Therefore, A is also complete.

(b) Since some of the eigenvalues are complex, the corresponding eigenvectors are also complex. Therefore, A does not admit an eigenbasis of $\mathbb{R}^3$. An eigenbasis of $\mathbb{R}^3$ must consist entirely of real eigenvectors, which do not exist for $\lambda_{2,3} = \pm i$.

(c) Yes, A admits an eigenbasis of $\mathbb{C}^3$. It consists of eigenvectors v_1, v_2, v_3 corresponding to $\lambda_1, \lambda_2, \lambda_3$:

$$\begin{aligned} \lambda_1 = 1: \quad & (A - I)v_1 = 0 \quad \Leftrightarrow \quad \begin{bmatrix} -1 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix} v_1 = 0 \quad \Leftrightarrow \quad v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}. \\ \lambda_2 = i: \quad & (A - iI)v_2 = 0 \quad \Leftrightarrow \quad \begin{bmatrix} -i & 0 & -1 \\ 0 & 1-i & 0 \\ 1 & 0 & -i \end{bmatrix} v_2 = 0 \quad \Leftrightarrow \quad v_2 = \begin{bmatrix} i \\ 0 \\ 1 \end{bmatrix}. \\ \lambda_3 = -i: \quad & v_3 = \overline{v_2} = \begin{bmatrix} -i \\ 0 \\ 1 \end{bmatrix}. \end{aligned} \tag{17}$$

Vectors v_1, v_2, and v_3 form an eigenbasis of $\mathbb{C}^3$, that is, a basis of $\mathbb{C}^3$ consisting of eigenvectors of A.

(d) As we know from Lecture 11, to construct a basis of $\mathbb{R}^3$ from the eigenbasis of $\mathbb{C}^3$, all we need to do is take the real and imaginary parts

$$v_1 = \mathrm{Re}(v_1), \quad v_2 = \mathrm{Re}(v_2) + i\mathrm{Im}(v_2), \quad v_3 = \overline{v_2} = \mathrm{Re}(v_2) - i\mathrm{Im}(v_2). \tag{18}$$

So

$$u_1 = v_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad u_2 = \mathrm{Re}(v_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad u_3 = \mathrm{Im}(v_2) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{19}$$

is a basis of $\mathbb{R}^3$, but it is not an eigenbasis, since u_2 and u_3 are not eigenvectors of A.

Problem 5. (10 points) Gershgorin Theorem
Consider the following matrix:

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \tag{20}$$

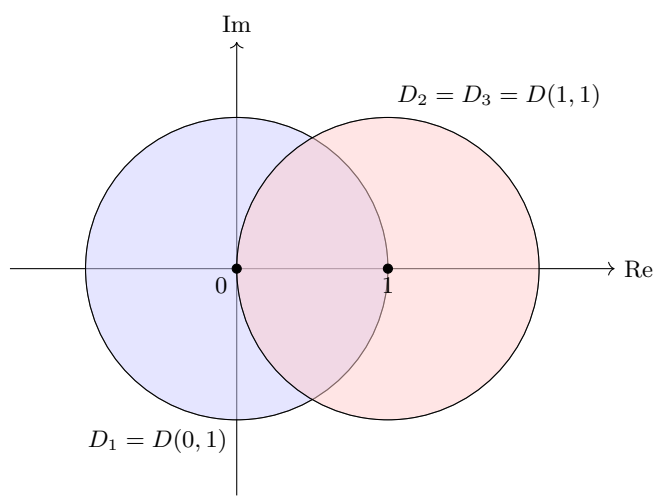
(a) (2 points) Find all Gershgorin disks and sketch the Gershgorin domain in the complex plane.
(b) (2 points) Show that the eigenvalues of any matrix A must lie in its *refined* Gershgorin domain, which is defined as the intersection of the Gershgorin domains of A and A^T, $D_A^* = D_A \cap D_{A^T}$.
(c) (2 points) Find and sketch the refined Gershgorin domain D_B^* of matrix B in (20).
(d) (2 points) Find the eigenvalues of B and check that they indeed belong to D_B^*.
(e) (2 points) Give an example of a nonsingular matrix A whose Gershgorin domain contains zero.

Solution:
(a) The Gershgorin disks D_i are

$$D_1 = D(0,1), \quad D_2 = D(1,1), \quad D_3 = D(1,1), \tag{21}$$

where $D(c,r)$ denotes the disk of radius r centered at $(c,0)$. The Gershgorin domain is

$$D_B = D_1 \cup D_2 \cup D_3 = D(0,1) \cup D(1,1), \tag{22}$$



(b) The eigenvalues of A and A^T are the same since the characteristic polynomials of the two matrices are identical:

$$p_{A^T}(\lambda) = \det(A^T - \lambda I) = \det(A - \lambda I)^T = \det(A - \lambda I) = p_A(\lambda). \tag{23}$$

Therefore, if $\lambda \in \operatorname{spec}(A)$, then, by the Gershgorin theorem, $\lambda \in D_A$. On the other hand, if $\lambda \in \operatorname{spec}(A)$, then $\lambda \in \operatorname{spec}(A^T)$, and, again by the Gershgorin theorem, $\lambda \in D_{A^T}$. This means that

$$\lambda \in \operatorname{spec}(A) \quad \Rightarrow \lambda \in D_A \cap D_{A^T}. \tag{24}$$

(c) The transpose of matrix B is

$$B^T = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \tag{25}$$

Its Gershgorin domain is

$$D_{B^T} = D(0,0) \cup D(1,2) \cup D(1,1) = D(1,2) \cup D(1,1). \tag{26}$$

Since $D_B \subset D_{B^T}$, the refined Gershgorin domain of B coincides with the original domain,

$$D_B^* = D_B \cap D_{B^T} = D_B = D(0,1) \cup D(1,1). \tag{27}$$

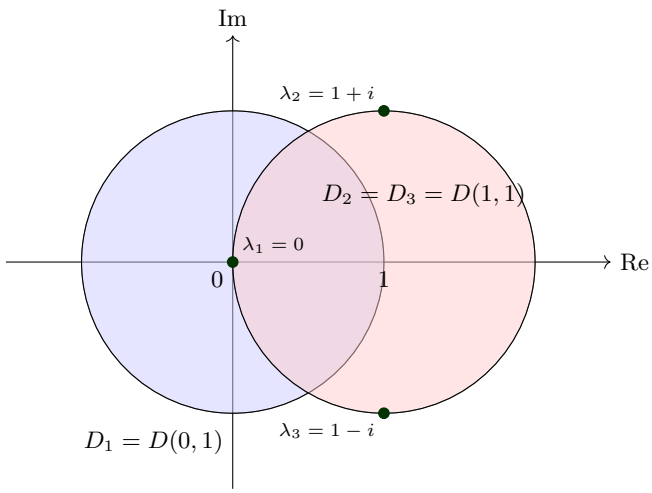
(d) The eigenvalues of B are the roots of the characteristic polynomial:

$$p_B(\lambda) = \det(B - \lambda I) = \det \begin{bmatrix} -\lambda & 1 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -1 & 1-\lambda \end{bmatrix} = -\lambda((1-\lambda)^2 + 1). \tag{28}$$

The eigenvalues are:

$$\lambda_1 = 0, \quad \lambda_2 = 1 + i, \quad \lambda_3 = 1 - i. \tag{29}$$

All eigenvalues belong to $D_B^* = D(0,1) \cup D(1,1)$.



(e) For example,

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{30}$$

This matrix is nonsingular, and its Gershgorin domain is the unit disk centered at zero.

Acknowledgements. A huge thanks to Pavlos Stavrinides for editing my clumsy MATLAB m-files and converting them into neat live scripts!

ACM/IDS 104 - Problem Set 5 - MATLAB Problems

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

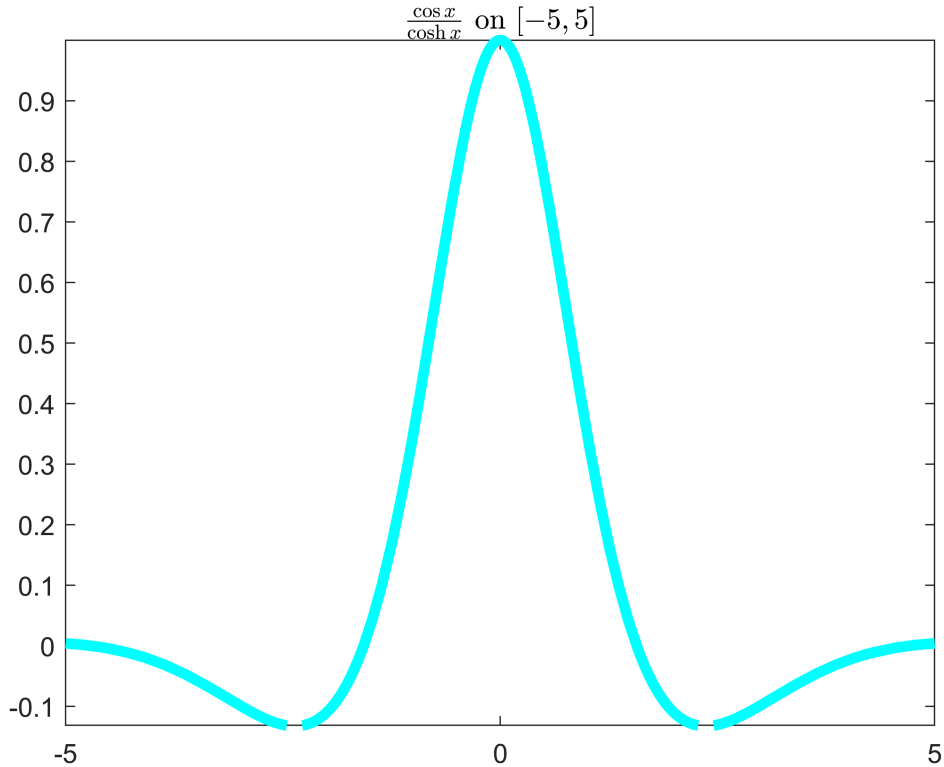
Problem 1 (10 points) Application of Projections to Approximation

In Problem 4 of PS4, we saw that even higher degree interpolating polynomials may not be accurate approximations to complex functions. We have the function:

$$f(x) = \frac{\cos x}{\cosh x}, \quad \text{on } [-a, a], \quad a = 5$$

Let us recall how this function looks like and how its interpolating polynomials of degree $(n-1)$ for $n = 3, 5, 10, 15$ behave:

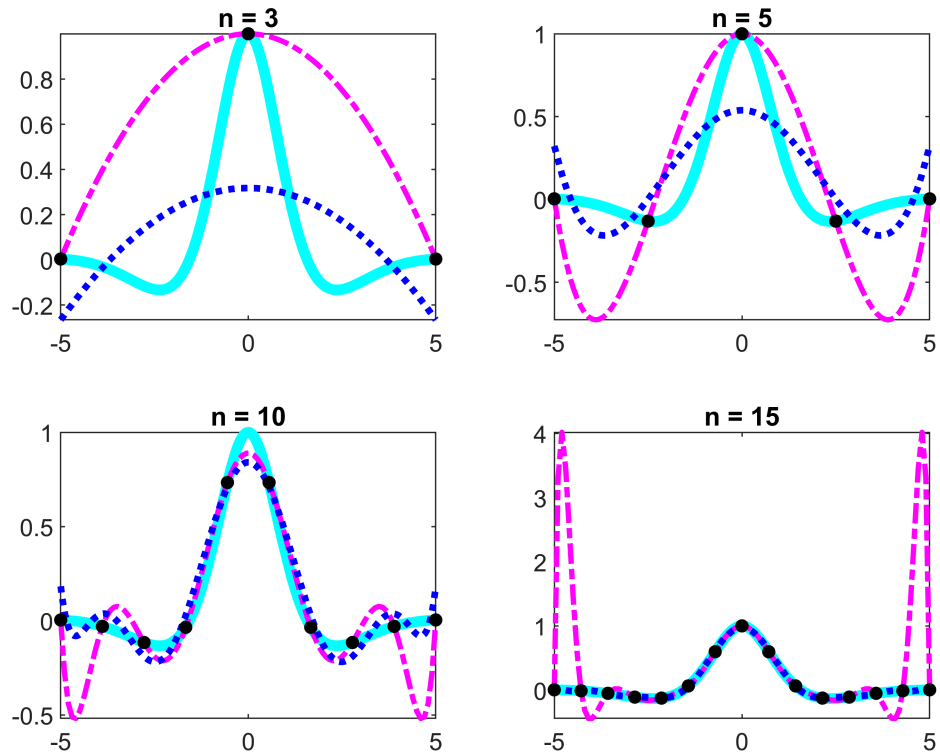
```
%{
Setup
%}
f = @(x) cos(x)./cosh(x); % our function
a = 5; % setting the value of a
n = [3 5 10 15]; % setting the number of points
sub = 1; % subplot index

%{
How f(x) looks like on [-5, 5]
%}
figure;
fplot(f, [-a, a], "-c", "lineWidth", 4);
title("$\frac{\cos{x}}{\cosh{x}}$ on $[-5, 5]$","Interpreter","latex");
```




```
%{
Read the discussion below and complete the code
%}
figure;
for ival = a
    for degree = n-1
        %{
        INTERPOLATING POLYNOMIALS -- no changes needed
        -> Select degree+1 points in the interval
        -> Evaluate f(x) on these points
        -> Find the polynomial coefficients
        %}
        pts = ones(degree+1, 2); % initializing the points
        pts(:, 1) = linspace(-ival, ival, degree+1); % setting the x-values
        for i = 1 : degree+1
            pts(i, 2) = f(pts(i, 1)); % evaluating cos(x) / cosh(x)
        end
        coeffs = polyfit(pts(:, 1), pts(:, 2), degree); % coefficients
        %{
        ORTHOGONAL PROJECTIONS -- TODO
        -> Get transformed Legendre polynomials
        -> Find alpha_k using L^2 inner product
        -> Evaluate alpha_k*Q_k
        %}
        bigsum = @(x) 0; %setting the sum
        for w = 1 : degree+1
            Q = @(q) legendreP(w - 1, q / ival); % getting Q_i transformed
```

```
            top = @(s) f(s) .* Q(s); % setting the numerator
            bottom = @(s) Q(s) .* Q(s); % setting the denomenator
            % obtaining alpha_i
            topgral = integral(top,-ival,ival); % evaluating numerator
            botgral = integral(bottom,-ival,ival); % evaluating denomenator
            alpha_i = topgral / botgral; % value of alpha_i
            alphaque = @(x) alpha_i * Q(x); % calcuating alpha_i * Q_i
            bigsum = @(x) bigsum(x) + alphaque(x); % updating the sum
        end
        %{
        PLOTTING
        Plot f(x), the sampled points, interpolating and approximating
        polynomials
        Please use different colors and linestyles
        %}
        subplot(2, 2, sub);
        fplot(f, [-ival, ival], "-c", "lineWidth", 4);
        hold on
        interpoints = linspace(-ival, ival);
        p = polyval(coeffs, interpoints); % evaluating coeffs in interval
        plot(interpoints, p, "-.m", "lineWidth", 2);
        fplot(bigsum, [-ival, ival] , ":b", "lineWidth", 2.5);
        plot(pts(:, 1), pts(:, 2), "ok", "MarkerSize", 2, "lineWidth", 3);
        title(strcat("n = ", int2str(degree+1)));
        sub = sub + 1; % increase subplot index
    end
end
```

```
Warning: Polynomial is badly conditioned. Add points with distinct X values, reduce the degree of the
 polynomial, or try centering and scaling as described in HELP POLYFIT.
```



Now, instead of interpolating polynomials, let us approximate $f(x)$ by its orthogonal projection onto the inner space $\mathscr{P}^{(n-1)}_{[-a,a]}$ of polynomials on $[-a,a]$, equipped with the L^2 inner product:

$$f(x) \approx p(x) = \mathrm{pr}_{\mathscr{P}^{(n-1)}_{[-a,a]}} f(x)$$

Recall (Lecture 10) that $p(x)$ is the closest (in the L^2 sense) polynomial to $f(x)$ in $\mathscr{P}^{(n-1)}_{[-a,a]}$, i.e.

$$p(x) = \arg \min_{q \in \mathscr{P}^{(n-1)}_{[-a,a]}} \|f(x) - q(x)\|$$

We know that the transformed Legendre polynomials $\widetilde{Q}_0(x), \cdots, \widetilde{Q}_{n-1}(x)$ form an orthogonal basis of $\mathscr{P}^{(n-1)}_{[-a,a]}$, and, therefore:

$$p(x) = \sum_{k=0}^{n-1} \alpha_k \widetilde{Q}_k(x)$$

where α_k are the coordinates of $p(x)$ in that basis.

Modify the above code to find the approximating polynomials as well. Plot each approximating polynomial on its corresponding subplot. Useful functions for this problem:

```
legendreP(), integral()
```